

Claire Shi Ye

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Education

Northwestern University, USA Ph.D. Astronomy, Advisor: Frederic A. Rasio	August 2022
Northwestern University, USA M.Sc. Physics, Advisor: Melville P. Ulmer	2016
Zhejiang University, China B.Sc. Physics · GPA: 3.82	2015

Honors & Awards

CITA Postdoctoral Fellowship	2022-present
Jeffrey L. Bishop Fellowship Bi-annually awarded to a CITA postdoc for outstanding research work: \$3k	2023-2025
IOP Publishing Top Cited Paper Award For the article 'On the Rate of Neutron Star Binary Mergers from Globular Clusters', Ye, C. S., et al. 2020, <i>ApJL</i> , 888, L10-22. One of the most cited papers from North America published across the entire IOP Publishing journal portfolio for 2020-2022. Top 1% of most cited articles in the Astronomy and Astrophysics subject category.	2023
IOP Publishing Top Cited Paper Award For the article 'Modeling Dense Star Clusters in the Milky Way and Beyond with the CMC Cluster Catalog', Kremer, K., Ye, C. S., et al. 2020, <i>ApJS</i> , 247, 48. One of the most cited papers from North America published across the entire IOP Publishing journal portfolio for 2020-2022. Top 1% of most cited articles in the Astronomy and Astrophysics subject category.	2023
NSF GK-12 'Reach for the Stars' Fellowship GK-12 Program · Collaborated with a K-12 science classroom teacher to bring more inquiry-based teaching methods into the classroom; Developed interactive Python programs for astronomy classroom activities; Co-organizer and instructor for the first CIERA/Northwestern high school astronomy summer camp	2018-2019

Grants & Proposals

An ultraviolet time-domain survey of the core-collapsed globular cluster NGC 6752 Co-I, Hubble Space Telescope Cycle 34, 2026	Awarded
Characterizing the dynamics of compact objects in the globular clusters 47 Tucanae, Terzan 5, and Liller 1 via N-body simulations Co-I, Hubble Space Telescope Cycle 33, Archival/Theory 2025	Awarded
A comprehensive view of the Milky Way's population of stellar-origin binaries Co-I, LISA Preparatory Science Proposal 2025	Awarded

Publications

51 total publications ([ADS Library](#)), including 10 first-author and 9 second-author (h-index=25).

FIRST- AND SECOND-AUTHOR PAPERS

19. **The Contribution of Disrupted Dense Star Clusters to Gaia's Compact Object Binaries**
Schiebelbein-Zwack, A. (graduate student), [Ye, C. S.](#), Reina-Campos, M. et al. 2026, [arXiv:2606.18419](#)
18. **Inferring Globular Cluster Initial Mass Function from Stellar Streams**
[Ye, C. S.](#) & Carlberg, R. G. 2026, [ApJL](#), 1007, L25

17. **Fast Radio Bursts from White Dwarf Binary Mergers: Isolated and Triple-Induced Channels**
Shariat, C. (graduate student), **Ye, C. S.**, Naoz, S., & Rose, S. 2026, [ApJL, 1000, L17](#)
16. **Mass Distribution of Binary Black Hole Mergers from Young and Old Dense Star Clusters**
Ye, C. S., Fishbach, M., Kremer, K., & Reina-Campos, M. 2026, [ApJ, 997, 267](#)
15. **Gravitational-wave kicks impact spins of black holes from hierarchical mergers**
Borchers, A. (graduate student), **Ye, C. S.**, & Fishbach, M. 2025, [ApJ, 987, 146](#)
14. **Predicting the rate of fast radio bursts in globular clusters from binary black hole observations**
Rao, A. (undergraduate student), **Ye, C. S.**, & Fishbach, M. 2025, [ApJL, 979, L12](#)
13. **Can slow pulsars in Milky Way globular clusters form via partial recycling?**
Kremer, K., **Ye, C. S.**, Heinke, C. O. et al. 2024, [ApJL, 977, L42](#)
12. **Lower-mass-gap Black Holes in Dense Star Clusters**
Ye, C. S., Kremer, K., Ransom, S. et al. 2024, [ApJ, 975, 77](#)
11. **The Redshift Evolution of the Binary Black Hole Mass Distribution from Dense Star Clusters**
Ye, C. S. & Fishbach, M. 2024, [ApJ, 967, 62](#)
10. **The Dominant Mechanism(s) for Populating the Outskirts of Star Clusters with Neutron Star Binaries**
Leigh, N. W., **Ye, C. S.**, Grondin, S. M. et al. 2024, [MNRAS, 527, 6913](#)
9. **Single Millisecond Pulsars from Dynamical Interaction Processes in Dense Star Clusters**
Ye, C. S., Kremer, K., Ransom, S. et al. 2024, [ApJ, 961, 98](#)
8. **On the Tidal Capture of White Dwarfs by Intermediate-mass Black Holes in Dense Stellar Environments**
Ye, C. S., Fragione, G., & Perna, R. 2023, [ApJ, 953, 141](#)
7. **Millisecond Pulsars in Dense Star Clusters: Evolution, Scaling Relations, and the Galactic-Center Gamma-ray Excess**
Ye, C. S. & Fragione, G. 2022, [ApJ, 940, 162](#)
6. **Formation of Low-mass Black Holes and Single Millisecond Pulsars in Globular Clusters**
Kremer, K., **Ye, C. S.**, Kiroğlu, F., et al. 2022, [ApJL, 934, L1](#)
5. **Compact Object Modeling in the Globular Cluster 47 Tucanae**
Ye, C. S., Kremer, K., Rodriguez, C. L., et al. 2022, [ApJ, 931, 84](#)
4. **Modeling Dense Star Clusters in the Milky Way and Beyond with the CMC Cluster Catalog**
(223 citations)
Kremer, K., **Ye, C. S.**, Rui, N. Z., et al. 2020, [ApJS, 247, 48-91](#)
3. **On the Rate of Neutron Star Binary Mergers from Globular Clusters** (166 citations)
Ye, C. S., Fong, W-f., Kremer, K., et al. 2020, [ApJL, 888, L10-22](#)
2. **Millisecond Pulsars and Black Holes in Globular Clusters** (102 citations)
Ye, C. S., Kremer, K., Chatterjee, S., et al. 2019, [ApJ, 877, 122-131](#)
1. **How Black Holes Shape Globular Clusters: Modeling NGC 3201** (79 citations)
Kremer, K., **Ye, C. S.**, Chatterjee, S., et al. 2018, [ApJL, 855, L15-21](#)

REFEREED CO-AUTHOR PAPERS

26. **Red vs. Blue: How metallicity shapes black hole dynamics and mergers in dense star clusters**
Agrawal, S., et al. (including **Ye, C. S.**) 2026, [arXiv:2606.23788](#)

25. **Black hole mergers from dense star clusters with realistic binary populations**
O'Connor, C. E., et al. (including Ye, C. S.) 2026, [arXiv:2606.14846](#)
24. **The diverse outcomes of binary white dwarf mergers and connections to Galactic LISA sources**
Kremer, K., Breivik, K., Ye, C. S. 2026, [arXiv:2605.05308](#)
23. **Very Massive Stars and High N/O: A Tale of the Nitrogen-enriched Super Star Cluster in the Sunburst Arc**
Shi, Y., Dai, L., Murray, N., Ye, C. S., et al. 2025, [ApJ](#), 997, 309
22. **Is GW231123 a hierarchical merger?**
Passenger, L., et al. (including Ye, C. S.) 2025, [ApJ](#), 999, 236
21. **The ejection and detectability of high- and hyper-velocity stars by compact object binaries in globular clusters**
Evans, F. A., Grondin, S. M., Ye, C. S., Webb, J., Laroche, A., Bovy, J. 2025, [arXiv:2506.14273](#)
20. **Connecting Cores and Black Hole Dynamics Across Scales: From Globular Clusters to Massive Ellipticals**
Kremer, K., et al. (including Ye, C. S.) 2025, [ApJL](#), 993, L34
19. **Binary properties of the globular cluster 47 Tuc (NGC 104). A dearth of short-period binaries**
Müller-Horn, J., et al. (including Ye, C. S.) 2025, [A&A](#), 693, A161
18. **Investigating cannibalistic millisecond pulsar binaries using MESA: new constraints from pulsar spin and mass evolution**
Misra, D., Linares, M., Ye, C. S. 2025, [A&A](#), 693, A314
17. **Gravitational Microlensing Rates in Milky Way Globular Clusters**
Kiroğlu, F., Weatherford, N., Kremer, K., Ye, C. S., et al. 2022, [ApJ](#), 928, 181
16. **Modeling Dense Star Clusters in the Milky Way and Beyond with the Cluster Monte Carlo Code**
Rodriguez, C. L., et al. (including Ye, C. S.) 2022, [ApJS](#), 258, 22
15. **White Dwarf Subsystems in Core-Collapsed Globular Clusters**
Kremer, K., et al. (including Ye, C. S.) 2021, [ApJ](#), 917, 28-46
14. **Matching Globular Cluster Models to Observations**
Rui, N. Z., et al. (including Ye, C. S.) 2021, [ApJ](#), 912, 102-118
13. **Fast Optical Transients from Stellar-Mass Black Hole Tidal Disruption Events in Young Star Clusters**
Kremer, K., Lu, W., Piro, A. L., Chatterjee, S., Rasio, F. A., Ye, C. S. 2021, [ApJ](#), 911, 104-116
12. **Intermediate-mass Black Holes from High Massive-star Binary Fractions in Young Star Clusters**
González, E., et al. (including Ye, C. S.) 2021, [ApJL](#), 908, L29-35
11. **Black Hole Mergers from Star Clusters with Top-Heavy Initial Mass Functions**
Weatherford, N. C., Fragione, G., Kremer, K., Chatterjee, S., Ye, C. S., et al. 2021, [ApJL](#), 907, L25-32
10. **Black Hole Mergers from Hierarchical Triples in Dense Star Clusters**
Martinez, M. A. S., et al. (including Ye, C. S.) 2020, [ApJ](#), 903, 67-83
9. **Populating the Upper Black Hole Mass Gap through Stellar Collisions in Young Star Clusters**
Kremer, K., et al. (including Ye, C. S.) 2020, [ApJ](#), 903, 45-62

8. **Demographics of Triple Systems in Dense Star Clusters**
Fragione, G., Martinez, M. A. S., Kremer, K., Chatterjee, S., Rodriguez, C. L., Ye, C. S., et al. 2020, [ApJ](#), 900, 16-38
7. **COSMIC Variance in Binary Population Synthesis**
Breivik, K., Coughlin, S. C., Zevin, M., Rodriguez, C. L., Kremer, K., Ye, C. S., et al. 2020, [ApJ](#), 898, 71-84
6. **GW190412 as a Third-generation Black Hole Merger from a Super Star Cluster**
Rodriguez, C. L., et al. (including Ye, C. S.) 2020, [ApJL](#), 896, L10-16
5. **The Optical Afterglow of GW170817: An Off-axis Structured Jet and Deep Constraints on a Globular Cluster Origin**
Fong, W.-f., et al. (including Ye, C. S.) 2019, [ApJL](#), 883, L1-9
4. **Black holes: The next generation-repeated mergers in dense star clusters and their gravitational-wave properties**
Rodriguez, C. L., et al. (including Ye, C. S.) 2019, [Phys. Rev. D](#), 100, 043027:1-15
3. **Post-Newtonian dynamics in dense star clusters: Binary black holes in the LISA band**
Kremer, K., et al. (including Ye, C. S.) 2019, [Phys. Rev. D](#), 99, 063003:1-12
2. **How Initial Size Governs Core Collapse in Globular Clusters**
Kremer, K., Chatterjee, S., Ye, C. S., et al. 2019, [ApJ](#), 871, 38-49
1. **Post-Newtonian dynamics in dense star clusters: Formation, masses, and merger rates of highly-eccentric black hole binaries**
Rodriguez, C. L., et al. (including Ye, C. S.) 2018, [Phys. Rev. D](#), 98, 123005:1-16

CONFERENCE PROCEEDINGS/RESEARCH NOTES

6. **No Black Holes in NGC 6397**
Rui, N. Z., et al. (including Ye, C. S.) 2021, [RNAAS](#), 5, 47
5. **The Observed Rate of Binary Black Hole Mergers can be Entirely Explained by Globular Clusters**
Rodriguez, C. L., et al. (including Ye, C. S.) 2021, [RNAAS](#), 5, 19
4. **The Role of “Black Hole Burning” in the Evolution of Dense Star Clusters**
Kremer, K., Ye, C. S., Chatterjee, S., et al. 2020, [IAU proceedings](#), 351, 357
3. **Shaping Si, NiCo, and glass substrates via stresses in the coatings**
Wang, X., Yao, Y., Ye, C. S., et al. 2016, [SPIE Conference Series](#), 9965, 99650D:1-9
2. **Toward large-area sub-arcsecond x-ray telescopes II**
O’Dell, S. L., et al. (including Ye, C. S.) 2016, [SPIE Conference Series](#), 9965, 996507:1-17
1. **APERTURE: a precise extremely large reflective telescope using re-configurable elements**
Ulmer, M. P., et al. (including Ye, C. S.) 2016, [SPIE Conference Series](#), 9904, 99041I:1-12

Presentations

45 total and 20 invited presentations. * marks invited talks

CONFERENCES

- | | |
|---|------------|
| 1. * Black Holes Across Mass Scales, Aspen Summer Program | June 2026 |
| 2. * CIFAR Gravity & the Extreme Universe Annual Meeting, King City, Canada | June 2026 |
| 3. * Max Planck Institute Group Leader Symposium, Berlin, Germany | March 2026 |

4. * CIERA Fellows at 15 Conference, Evanston, IL	August 2025
5. * IAU Symposium 398 & MODEST-25, Seoul, Korea	June 2025
6. * EAS Annual Meeting (special session 'Compact Binary Pulsars')	June 2025
7. * 55th AAS Division of Dynamical Astronomy Meeting, Atlanta, GA	May 2025
8. Globular Clusters and Their Tidal Tails Conference, Toronto, Canada	May 2024
9. 54th AAS Division of Dynamical Astronomy Meeting, Toronto, Canada	May 2024
10. Intermediate-mass Black Holes Meeting II, San Pedro, Belize	December 2023
11. MODEST-23, Evanston, IL	August 2023
12. Intermediate-mass Black Holes Meeting I, San Juan, PR	May 2022
13. AAS HEAD Meeting (stellar/compact object session), Pittsburgh, PA	March 2022
14. Dynamical Formation of Gravitational Wave Sources, Aspen, CO	January 2022
15. * 16th Marcel Grossmann Meeting	July 2021
16. EAS Annual Meeting (session 'Where are the BH-NS binaries')	June 2021
17. IAU Symposium 351 & MODEST-19, Bologna, Italy	May 2019
18. Midwest Relativity Meeting, Milwaukee, WI	October 2018
19. MODEST-18, Santorini, Greece	June 2018

SEMINARS AND COLLOQUIUMS

1. * Texas Tech University Physics & Astronomy Colloquium	February 2026
2. * Perimeter Institute Colloquium	October 2025
3. * University of Guelph Department of Physics Colloquium	October 2025
4. * CCA Star & Plasma Astrophysics Meeting	May 2025
5. * Michigan State University Department of Physics and Astronomy Colloquium	April 2025
6. * Caltech TAPIR Seminar	April 2025
7. * UC San Diego Simulation and Theory Meeting	March 2025
8. * Saint Mary's University Astronomy Colloquium	March 2025
9. Columbia University THEA Seminar	November 2024
10. Princeton University Thunch Talk	November 2024
11. UC Berkeley Explosive Astro Seminar	November 2024
12. * UCLA TEPAPP Seminar	November 2024
13. Carnegie Mellon University Multi-messenger Astronomy Seminar	October 2024
14. * Perimeter Institute Strong Gravity Seminar	November 2023
15. * McMaster University Astro Group Talk	May 2023
16. * McGill Space Institute Seminar	February 2023
17. UC-Santa Cruz FLASH Seminar	December 2021
18. Carnegie Observatories Lunch Talk	November 2021
19. UCLA Lunch Talk	November 2021
20. Caltech TAPIR Seminar	November 2021
21. Princeton University Galread Seminar	October 2021
22. Columbia University Astro Seminar	October 2021
23. Carnegie Mellon University & University of Pittsburgh Astro Lunch Seminar	October 2021
24. National Radio Astronomy Observatory TUNA Lunch Talk	May 2021
25. Texas Tech University Summer Astro Seminar	May 2021
26. CCA Stars & Compact Objects Meeting	May 2021

27. Northwestern University Brown Bag Seminar

April 2019

Leadership & Professional Service

Cluster Monte Carlo Collaboration

Develop and implement compact object physics in the state-of-the-art Cluster Monte Carlo N -body code; lead development of pulsar models (magnetic fields, spin periods) and neutron star/low-mass black hole dynamics (tidal capture interactions, tidal disruption and spin-up).

COSMIC Collaboration

Develop and implement physics for pulsars (magnetic fields and spin periods) and white dwarf mass transfer in the rapid population synthesis code COSMIC.

Conference/Workshop Organizer

- CITA Postdoc Advance Workshop: Annual workshops focused on skill development for CITA/CITA National fellows across Canada 2022-present
- Globular Clusters and Their Tidal Tails–From the Milky Way to the Local Group: Week-long conference Toronto 2024
- CIERA Pulsar Workshop: A three-day workshop of pulsar physics and dynamics in globular clusters Northwestern 2019

Peer Reviewer

The Astrophysical Journal, the Astrophysical Journal Letters, the Monthly Notices of the Royal Astronomical Society, Nature, Nature Astronomy, and the Open Journal of Astrophysics

Outreach/Departmental Service

DEPARTMENTAL SERVICE

Northwestern Physics and Astronomy Graduate Student Council

2021-2022

Master's Student Committee Chair · Supported Master's students by ensuring awareness of policies, deadlines, and other information pertinent to Master's students success both at Northwestern and in the future, and provided the department with authentic feedback as the Master's program develops from its infancy

CIERA K-12 Task Force

Northwestern 2021-2022

Committee · Developed a framework for creating and sustaining K-12 outreach initiatives at CIERA with the goal of ensuring that CIERA K-12 outreach has a social justice impact

OUTREACH

CIERA Astronomer Evening

2018-2022

Monthly conversations with the public in Dearborn Observatory including open Q & A sessions and interactive activities

Astronomy on Tap Chicago

2017-2022

Engaged the public at local venues with professional astronomy talks, trivia, and prizes once per quarter as part of a national outreach effort

Letters to a Pre-Scientist

2018-2019

Exchanged letters with middle school students in high-poverty areas to demystify STEM career and inspire future scientists

Northwestern Seven Minutes of Science

2017

TED-style public symposium on [Pulsars in the Snow Globes](#)

Helix Magazine

2017

Outreach article on the story of two camps of astronomers behind the discovery of the first binary black hole merger: [Astronomy Fugato: Two Approaches, One Vast Field of Discovery](#)

Teaching Experience

Northwestern University

2016-2017 & Fall 2020

Teaching assistant · Taught weekly discussions or lab sessions for four different undergraduate General/College Physics courses · Taught a graduate course on Methods of Theoretical Physics

CIERA High School Summer Camp

2019

Co-organizer · Co-organized the high school summer camp with team-style learning, hands-on training, real astronomy research experiences and introductory lectures
Lecturer · Taught multiple lectures ranging from astronomy to computer programming

Niles North High School

2018-2019

Teaching assistant in astronomy classes · GK-12 Program

Student Mentoring

Aryamann Rao (one publication)

University of Toronto 2023-2025

Undergraduate project: Predicting the rate of fast radio bursts in globular clusters from binary black hole observations

Angela Borchers (one publication)

Max Planck Institute 2024-2025

PhD project: Gravitational-wave kicks impact spins of black holes from hierarchical mergers

Cheyenne Shariat (one publication)

Caltech 2025-2026

PhD project: Fast Radio Bursts from White Dwarf Binary Mergers: Isolated and Triple-induced Channels

Aryanna Schiebelbein-Zwack (one paper submitted)

CITA 2024-present

PhD project: Gaia binaries from dissolved star clusters

Mark Dodici (two papers in preparation)

University of Toronto 2025 spring-present

PhD project: Modeling Intermediate-mass black holes in dense star clusters

Aeysha Munawwarah

University of Toronto 2025 summer-present

CITA Summer Undergraduate Research Fellowship and senior project: The effects of massive star binary properties on black hole contents in dense star clusters

Dang Pham

University of Toronto 2023-2024

University of Toronto postdoc-graduate student mentorship program

Steffani Grondin

University of Toronto 2022-2023

University of Toronto postdoc-graduate student mentorship program

Rachel Zhang

Northwestern University 2022-2024

PhD project: Studying the effects of dynamics on idealized binary populations

Workshops

Black Holes Across Mass Scales (competitive)

Aspen 2026

Summer program at the Aspen Center for Physics

Stellar-mass Black Holes at the Nexus of Optical, X-ray, and Gravitational Wave Surveys (competitive)

Santa Barbara 2025

KITP program

Dynamical Formation of Multi-band Transients in Galactic Nuclei

Santa Barbara 2025

One-week KITP program

Dynamical Formation of Transients in Galactic Nuclei (competitive)

Aspen 2024

Two-week working group at the Aspen Center for Physics

4th POSYDON Collaboration Meeting (invitation-only)

Geneva 2024

One-week meeting of collaborative hacking and strategic discussions on the next-generation binary population synthesis code POSYDON

Stellar Interactions and the Transients They Cause (competitive)

Aspen 2023

Summer program at the Aspen Center for Physics

Black Hole Dynamics in Clusters

Northwestern 2018

One-week workshop on black hole dynamics

Heidelberg Summer School

University of Heidelberg 2017

One-week summer school on compact objects & gravitational waves

MESA Summer School

UC Santa Barbara 2017

One-week summer school on the stellar evolution code MESA

Research Communication Training Program

Northwestern 2017

Ten-week courses on science communication and presentation skills, culminated in a TED-style presentation